Mauro Artigiani

Personal Data

Place of Birth:	Rome, Italy
Nationality:	Italian
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Academic Positions

2024–	Profesor Asociado, **Departamento de Matemáticas**, Universidad Nacional de Colombia, Sede Bogotá, Colombia.
2022–2024	Graduate School Director, **School of Engineering, Science and Technology**, Universidad del Rosario, Bogotá, Colombia.
2020–2024	Profesor Principal, **School of Engineering, Science and Technology**, Universidad del Rosario, Bogotá, Colombia.
2018–2019	Investigador postdoctoral, **Departamento de Matemáticas**, Universidad de los Andes, Bogotá, Colombia.
2016–2018	Postdoctoral researcher, **Centro di Ricerca Matematica Ennio De Giorgi**, Scuola Normale Superiore, Pisa, Italy.

Education

2012–2016	Ph. D. in Mathematics, **University of Bristol**, Bristol, UK. Thesis: *"Some aspects of Diophantine approximation and ergodic theory of translation surfaces"*. Advisor: Prof. Corinna Ulcigrai.
2010–2012	Master of Science in Mathematics **University of Rome "Sapienza"**, Rome. 110/110 *cum laude* | Major: Algebra & Geometry. Thesis: *"Flat surfaces and the Veech dichotomy"*. Advisor: Prof. Gabriele Mondello.
2007–2010	Bachelor of Science in Mathematics **University of Rome "Sapienza"**, Rome. 110/110 *cum laude* | Major: Pure Mathematics. Thesis: *"On conformal structures of surfaces"*. Advisor: Prof. Andrea Sambusetti.

Grants and Awards

2021	Beca de Estancia de Docencia e Investigación 2021-II.
2021	Invited Professor, University of Pisa.
2019	Structured Quartet Research Ensemble (SQuaRE), American Institute for Mathematics, *joint with* Anja Randecker, Chandrika Sadanand, Ferrán Valdez and Gabriela Weitze-Schmithüsen
2012–2016	University of Bristol postgraduate Scholarship
2011–2012	Student in the **"Master Excellence Path"** in Mathematics at the University of Rome "Sapienza".
2008–2010	Student in the **"Excellence Path"** in Mathematics at the University of Rome "Sapienza".

Publications and Preprints

1. Typical Weak Mixing and Exceptional Spectral Properties for Interval Translation Mappings. *joint with* Artur Avila, Sébastien Ferenczi, Pascal Hubert *and* Alexandra Skripchenko. *preprint*. arXiv

2. Veech groups of covers of the Chamanara surface.
joint with Anja Randecker, Chandrika Sadanand, Ferrán Valdez *and* Gabriela Weitze-Schmithüsen. *preprint*. arXiv.

3. Renormalization for Bruin-Troubetzkoy ITMs.
joint with Pascal Hubert *and* Alexandra Skripchenko. *Discrete and Continuous Dynamical Systems*, **47** (2026), 519–547. Journal; arXiv.

4. Realizing groups as symmetries of infinite translation surfaces.
joint with Anja Randecker, Chandrika Sadanand, Ferrán Valdez *and* Gabriela Weitze-Schmithüsen. To appear in *Algebraic & Geometric Topology*. arXiv.

5. On rigidity properties of time-changes of unipotent flows.
joint with Livio Flaminio *and* Davide Ravotti. To appear in *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*. Preliminary journal version. arXiv.

6. A note on double rotations of infinite type.
joint with Charles Fougeron, Pascal Hubert *and* Alexandra Skripchenko. *Transactions of the Moscow Mathematical Society*, **82** (2021), 157–172. Journal; arXiv.

7. Octagonal continued fraction and diagonal changes.
Rendiconti dell'Istituto di Matematica dell'Università di Trieste, **53** (2021), N. 9, 1–28. Journal; arXiv

8. Persistent Hall rays for Lagrange spectra at cusps of Riemann surfaces.
joint with Luca Marchese *and* Corinna Ulcigrai. *Ergodic theory and Dynamical systems*, **40** (2020), 2017–2072. Journal; arXiv.

9. Exceptional ergodic directions in Eaton lenses.
Israel Journal of Mathematics, **220** (2017), 29–56. Journal; arXiv.

10. The Lagrange spectrum of a Veech surface has a Hall ray.
joint with Luca Marchese *and* Corinna Ulcigrai. *Groups, Geometry, and Dynamics*, **10** (2016), 1287–1337. Journal; arXiv.

Talks

October 2024	*Dynamical Systems Seminar*, Higher School of Economics, Moscow, Russia.
October 2024	*Geometry and Topology Seminar*, Higher School of Economics, Moscow, Russia.
June 2024	*Dynamical Systems Seminar*, CRM Ennio De Giorgi, Scuola Normale Superiore, Pisa, Italy.
November 2023	*Colloquium talk*, Semana de las Ciencias, Universidad Distrital Francisco José de Caldas, Bogotá, Colombia.
October 2023	*Dynamical Systems Seminar*, CRM Ennio De Giorgi, Scuola Normale Superiore, Pisa, Italy.
June 2023	*Semi-plenary talk*, XXIII Congreso Colombiano de Matemáticas, UPTC, Tunja, Colombia.
December 2022	*Combinatorial and Geometric Approaches on Dynamics*, Centro de Modelamiento Matemático, Universidad de Chile, Santiago, Chile.
October 2022	*Colloquium talk*, Universidad de Los Andes, Bogotá, Colombia.
June 2022	*Courant Institute Dynamical Systems Seminar*, New York University, USA.
June 2021	*Ergodic Theory Workshop*, University of North Carolina at Chapel Hill, USA.
June 2021	*DinAmicI*, Italy, recording.
March 2021	*Colloquium talk*, Universidad Nacional de Colombia, Bogotá, Colombia.
September 2019	*Colloquium talk*, Pontificia Universidad Javeriana, Bogotá, Colombia.
September 2019	*Colloquium talk*, Universidad Nacional de Colombia, Bogotá, Colombia.
June 2019	*Dynamical Systems Seminar*, CRM Ennio De Giorgi, Scuola Normale Superiore, Pisa, Italy.
October 2018	*Colloquium talk*, Universidad de Los Andes, Bogotá, Colombia.
February 2018	*Workshop Around Dynamical Systems*, Udine, Italy.
December 2017	*Dynamical Systems Seminar*, Institut de Mathématiques de Marseille, France.

Talks (continued)

November 2017	*Horowitz Seminar*, Tel Aviv University, Israel.
October 2017	*Colloquium talk*, Universidad de Los Andes, Bogotá, Colombia.
August 2017	*Dynamics in Number Theory and Geometry*, Fields Institute, Kingston, Canada.
June 2017	*Workshop on Moduli Spaces*, SkolTech & HSE, Moscow, Russia.
November 2016	*Prima Giornata DinAmica*. Gran Sasso Science Institute, L'Aquila, Italy.
February 2016	*Seminar at the Abdus Salam International Centre for Theoretical Physics*. Trieste, Italy.
November 2015	*Simons workshop on Translation surfaces and Dynamics*. Bedlewo, Poland.
July 2015	*School: "Dynamics & Geometry in Teichmüller Space"*. Marseille, France. *(contributed)*
June 2015	*DinAmici IV*, conference of young Italian dynamicist. Corinaldo, Italy.
February 2015	*Ergodic Theory and Dynamical Systems seminar*. Warwick University, UK.
October 2014	*Ergodic Theory and Dynamical Systems seminar*. University of Bristol, UK. *(contributed)*

Teaching

2024-II	Fundamentos de Matemáticas, Variable Compleja y Matemáticas Básicas.
2024-I	Teoría de la Computación and Álgebra Abstracta y Codificación.
2023-II	Teoría de la Computación and Álgebra Abstracta y Codificación.
2023-I	Teoría de la Computación and Álgebra Abstracta y Codificación.
2022-II	Teoría de la Computación and Álgebra Abstracta y Codificación.
2022-I	Lógica, Teoría de Números y Conjuntos, Álgebra Abstracta y Codificación and Geometría Diferencial y Computacional.
2021-II	Teoría de la Computación, and Geometría Diferencial y Computacional.
June 2021	Graduate course on Translation Surfaces, Ph.D. program University of Pisa.
2021-I	Lógica, Teoría de Números y Conjuntos, and Cálculo 2.
2020-II	Teoría de la Computación, and Álgebra Abstracta y Codificación.
2020-I	Lógica, Teoría de Números y Conjuntos, and Álgebra Abstracta y Codificación.
2019-II	Cálculo diferencial and Teoría Ergódica.
2019-I	Cálculo integral con ecuaciones diferenciales and Introducción a sistemas dinámicos.
2018-II	Cálculo integral con ecuaciones diferenciales.
2017–2018	Dynamical systems & Ergodic Theory *(shared)*.
2014–2016	Pure Mathematics subjects first year tutorials.
2013	Marker for "Ergodic Theory and Dynamical Systems".

Advising

Juliana Bermúdez Guzmán	Undergraduate student, Universidad del Rosario, 2024–.
Daniel Pallejá López	Master student, Universidad de los Andes, 2018–2020.

Conferences and Meeting Organization

2019	Weekly Colloquium Seminar, Universidad de los Andes, Bogotá, Colombia.
2018	From order to chaos, CRM Ennio De Giorgi, Pisa.
2017–2018	Dynamical Systems seminar, with Paolo Giulietti.
2014	*MINGLE*, University of Bristol, UK. MINGLE is a one day meeting to welcome the new PhD and Master students at the University of Bristol. I was one of the five organisers and approximately 50 people were present.

Outreach

Google, las Matemáticas, y los Sistemas Dinámicos (YouTube video interview)

Service

Reviewer for zbMATH Open.

Languages

Italian:	Native
English:	Fluent
Spanish:	Fluent